

Notes on Category Theory

from Hilton and Wu, A Course in Modern Algebra

Notations

$\mathfrak{C}, \mathfrak{D}$: categories

F, G : functors

$\text{Ob}(\mathfrak{C})$: the set of objects of category $\mathfrak{C}$

$\text{Mor}(\mathfrak{C})$: the set of morphisms of category $\mathfrak{C}$

$\mathfrak{C}(X, Y)$: the set of morphisms from object X to object Y

$\mathfrak{S}$: the category of sets

$\mathfrak{G}$: the category of groups

$\mathfrak{T}$: the category of topological spaces and continuous functions

$\mathfrak{Ab}$: the category of abelian groups

$\mathfrak{D}_K$: the category of finite dimensional vector spaces over the field K

Categories

Axioms of Categories:

CAT-1 The sets $\mathfrak{C}(A, B)$ and $\mathfrak{C}(C, D)$ are disjoint unless $(A, B) = (C, D)$.

CAT-2 $f : A \rightarrow B, g : B \rightarrow C, h : C \rightarrow D$, then $h(gf) = (hg)f$

CAT-3 For any object A , there is an identity morphism $1_A : A \rightarrow A$ such that for any other two morphisms $f : A \rightarrow B, g : C \rightarrow A, f1_A = f, 1_Ag = g$.

Definition 1 (Initial and Terminal Objects): An object I is said to be *initial*, if the set of morphisms $\mathfrak{C}(I, -)$ is a singleton. An object T is said to be *terminal (coinitial)* if the set of morphisms $\mathfrak{C}(-, T)$ is a singleton, where $-$ is any object in $\mathfrak{C}$. If an object is both initial and terminal it is called a zero.

Example: $\{\}$ is the initial object in $\mathfrak{S}$. 1 is the zero object in $\mathfrak{G}$.

Remark: A zero object is also called *universal*.

Functors

Definition 2 (Forgetful functor): A functor $U : \mathfrak{C} \rightarrow \mathfrak{S}$ where $\mathfrak{S}$ is the category of the underlying set of $\mathfrak{C}$ and $U(f)$ is a function in $\mathfrak{S}$ such that $f(X) = U(f)(U(X))$. For example, $U : \mathfrak{D}_K \rightarrow \mathfrak{Ab}$ is a forgetful functor "forgetting" the multiplication operation in $\mathfrak{D}_K$.

Definition 3 (Faithful and Full Functors): A functor $F : \mathfrak{C} \rightarrow \mathfrak{D}$ induces a map $F^* : \mathfrak{C}(X, Y) \rightarrow \mathfrak{D}(FX, FY)$. A faithful functor F (resp. full functor F) induces an injective (resp. surjective) map F^* for all X, Y .

Example: The forgetful functor $U : \mathfrak{G} \rightarrow \mathfrak{S}$ is faithful because all homomorphisms are functions between the sets underlying a group and the map $\varphi \mapsto U(\varphi)$ is injective. However, it is not full.

Natural Transformations

Definition 4 (Natural Transformations): A map $\tau : F \rightarrow G$ is called a natural transformation if it assigns to each object $X \in \mathfrak{C}$ a morphism τ_X such that the following diagram commutes:

$$\begin{array}{ccc}
FX & \xrightarrow{\tau_X} & GX \\
F(f) \downarrow & & \downarrow G(f) \\
FY & \xrightarrow{\tau_Y} & GY
\end{array}$$

Example: In $\mathfrak{D}_K$, we have

$$\begin{array}{ccc}
V & \xrightarrow{v \mapsto \tilde{v}} & V^{**} \\
f \downarrow & & \downarrow f^{**} \\
W & \xrightarrow{w \mapsto \tilde{w}} & W^{**}
\end{array}$$

where $v \mapsto \tilde{v}$ is the isomorphism between V and the double dual space V^{**} and $(f^{**}(v^{**}))(\phi) = v^{**}(f^*(\phi))$ and f^* is the dual map of f and $\phi \in W^*$. Here $\tau : \text{id} \rightarrow (* *)$.

Duality Principle

We say the category $\mathfrak{C}^{\text{op}}$ *the opposite category of $\mathfrak{C}$* if $\text{Ob}(\mathfrak{C}) = \text{Ob}(\mathfrak{C}^{\text{op}})$ and $\mathfrak{C}(X, Y) = \mathfrak{C}^{\text{op}}(Y, X)$ such that $f \circ^{\text{op}} g = g \circ f$.

An example of dual statment S : $S(\mathfrak{C})$: I is an initial object if $\mathfrak{C}(I, X)$ is a singleton for every $X \in \mathfrak{C}$.

$S(\mathfrak{C}^{\text{op}})$: I is an initial object if $\mathfrak{C}^{\text{op}}(I, X)$ is a singleton for every $X \in \mathfrak{C}^{\text{op}}$

$S^{\text{op}}(\mathfrak{C})$: I is a coinital object if $\mathfrak{C}(X, I)$ is a singleton for every $X \in \mathfrak{C}^{\text{op}}$.

If S is a statement such that it is meaningful in all categories. If S is true in $\mathfrak{C}$, then S is also true in $\mathfrak{C}^{\text{op}}$